

Ishtier Rahman

M.Eng., EIT, CSWA, LEED Green Associate

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Summary

Mechanical Engineer (EIT) with 4+ years of combined experience in mechanical design, industrial equipment, product development, and maintenance engineering. Experienced in industrial combustion systems, piping layouts, valve racks, mechanical components, and manufacturing documentation using Bentley MicroStation and SOLIDWORKS. Skilled in preparing mechanical drawings, P&IDs, technical specifications, and engineering documentation while supporting design analysis, prototype testing, commissioning, and continuous product improvement. Proven ability to collaborate with multidisciplinary engineering, manufacturing, and project teams to deliver practical and manufacturable mechanical solutions.

Areas of Expertise

- ♦ Mechanical System & Equipment Design
- ♦ Product Development & Design Engineering
- ♦ Industrial Piping & Valve Rack Design
- ♦ Combustion Systems & Process Equipment
- ♦ 3D CAD Modeling & 2D Drafting
- ♦ Manufacturing Drawings & Technical Documentation
- ♦ Component Selection & Engineering Specifications
- ♦ P&IDs, Schematics & BOM Preparation
- ♦ Prototype Testing & Design Verification
- ♦ Commissioning & Cross-Functional Engineering

Technical Skills

- **CAD & Design:** Bentley MicroStation, SOLIDWORKS (CSWA), AutoCAD
- **Mechanical Design:** Industrial Piping, Valve Racks, Combustion Systems, Equipment & Component Selection
- **Simulation & Analysis:** ANSYS Fluent (CFD), MATLAB, Minitab
- **Engineering Documentation:** P&IDs, Manufacturing Drawings, BOMs, Schematics, Technical Specifications
- **Energy & Engineering Tools:** RETScreen, System Advisor Model (SAM)
- **Programming & Software:** Python, SAP, LaTeX, Microsoft Office

Work Experience

A.H. Burns Energy Systems Ltd, Ancaster, ON

07/2026 – Present

Mechanical Engineer in Training

- Design and develop mechanical systems and components for industrial combustion equipment used in zinc oxide production.
- Create detailed mechanical drawings, schematics, and manufacturing documentation using Bentley MicroStation and other CAD software.
- Support the design of industrial piping systems, including air and natural gas pipework, valve racks, combustion control systems, and structural components.
- Perform engineering analysis and design evaluations to improve system performance, reliability, and manufacturability while collaborating with electrical, software, manufacturing, and project engineering teams.
- Select and evaluate mechanical components, piping fittings, valves, regulators, pressure switches, and instrumentation based on operating requirements and project specifications.
- Prepare technical documentation and manufacturing packages and support prototype development, in-house testing, design verification, commissioning, troubleshooting, and field support for domestic and international projects.

US Air Compressor, Burlington, ON

05/2025 – 07/2026

Maintenance and Technical Support Engineer

- Provided engineering support for industrial compressed air systems, including troubleshooting, performance validation, and system optimization across manufacturing facilities.
- Supported retrofit and installation projects by updating system layouts and component configurations in SOLIDWORKS/AutoCAD and maintaining accurate schematics and P&IDs.
- Assisted with equipment sizing and specifications for compressors, dryers, filters, receivers, and controls to meet performance and compliance requirements.

- Supported commissioning and startup verification, resolving integration and performance issues during field validation and system upgrades.
- Developed technical documentation, procedures, specifications, and drawings while contributing to energy optimization initiatives that delivered estimated annual client savings of \$20K–\$30K.

Wellwise Home Health & Wellness, London, ON
Customer Service & Technical Support Associate

09/2024 – 05/2025

- Advised customers on product selection and provided basic troubleshooting support for home medical equipment.
- Supported order processing, inventory coordination, and service documentation, improving workflow efficiency.
- Collaborated with service teams to resolve equipment-related issues efficiently while maintaining customer service and quality standards.

Carleton University, Ottawa, ON
Graduate Teaching Assistant

09/2021 – 04/2023

- Facilitated engineering labs and tutorials for 300+ students in thermodynamics, heat transfer, HVAC, and related mechanical engineering subjects.
- Prepared course materials, supervised laboratory activities, and provided technical support and feedback to undergraduate students.

AVOS Ltd., Dhaka, Bangladesh
Maintenance Engineer

03/2019 – 07/2021

- Increased equipment performance by 25% across 50 machinery systems through systematic inspections, troubleshooting, and maintenance optimization.
- Reduced unplanned downtime by 15%, saving approximately 500K BDT annually, through preventive maintenance improvements.
- Directed installation and commissioning of 30+ machines valued at over 1.2 million BDT, ensuring successful integration and reliable operation.
- Supported engineering modifications, component changes, and technical documentation to improve equipment performance and reduce repeat failures.

Selected Engineering & Simulation Projects

Computational Analysis of Wind Flow Over Building Rooftops

2018 – 2019

- Conducted CFD simulations using ANSYS Fluent to analyze wind flow over various rooftop geometries (dome, filleted, half-wedge, convergent, tulip).
- Assessed velocity profiles and contours to evaluate the impact of geometry on air movement and turbine placement.
- Compared geometric performance and identified dome/filleted profiles as providing enhanced wind speed amplification over conventional designs.

Numerical Analysis of Flow & Heat Transfer Between Parallel Plates with Constrictions

2017 – 2018

- Developed CFD models to study flow acceleration, turbulence, and convective heat transfer with varying geometry.
- Evaluated pressure, velocity, temperature, heat transfer coefficient, and Nusselt number across multiple design cases.
- Performed parametric studies by changing block size/quantity and analyzed recirculation effects on thermal performance.

Education

Master of Engineering in Mechanical Engineering – Carleton University, Ottawa, ON, Canada (2021-2024)

Master of Science in Renewable Energy Technology – University of Dhaka, Dhaka, Bangladesh (2018-2021)

Bachelor of Science in Mechanical Engineering – Military Institute of Science and Technology, Dhaka, Bangladesh (2014-2018)

Professional Certifications & Training

- Engineer in Training (EIT), EIBC
- LEED Green Associate, GBCI
- Certified Associate in Project Management (CAPM), PMI
- Certified SOLIDWORKS User (CSWA, Sustainability, Additive Manufacturing), Dassault Systèmes
- CSA Certification Training (CSA C22.2 No. 68 & SPE-1000), US Air Compressor
- Lean Six Sigma Yellow Belt Certification, SSGI